7) $y'' - 5y' + 6y = e^{3x} + 12x - 4$

a) $\lambda^2 - 5\lambda + 6 = 0$

$\lambda_{1,2} = \{2, 3\} \rightarrow y_h(x) = a e^{2x} + b e^{3x}$

b) $b(x) = e^{3x} + 12x - 4$

$e^{3x} \xrightarrow{\text{zákl.}} Ae^{3x} \xrightarrow{\lambda=3} \text{korekce} \rightarrow xAe^{3x}$

$12x - 4 \rightarrow Bx + C \xrightarrow{\lambda=0} \text{bez kor.} \rightarrow Bx + C$

$y_p = xAe^{3x} + Bx + C$

$L = [xAe^{3x} + Bx + C]'' - 5[xAe^{3x} + Bx + C]' + 6[xAe^{3x} + Bx + C] =$

$= [\quad] - 5[Ae^{3x} + 3xAe^{3x} + B] + 6[xAe^{3x} + Bx + C] =$

$9xAe^{3x} + 6Ae^{3x}$

$= 9xAe^{3x} + 6Ae^{3x} - 5Ae^{3x} - 15xAe^{3x} - 5B + 6xAe^{3x} + 6Bx + 6C =$

$= 0xAe^{3x} + 1Ae^{3x} + 6Bx + (-5B + 6C) = Ae^{3x} + 6Bx + (-5B + 6C) = e^{3x} + 12x - 4$

$\begin{bmatrix} A = 1 \\ 6B = 12 \\ (6C - 5B) = (-4) \end{bmatrix} \rightarrow \begin{matrix} A = 1 \\ B = 2 \\ C = 1 \end{matrix} \rightarrow y_p(x) = xe^{3x} + 2x + 1$

$y(x) = ae^{2x} + be^{3x} + xe^{3x} + 2x + 1 \; ; x \in \mathbb{R}$

$y'(x) = 2ae^{2x} + 3be^{3x} + 3xe^{3x} + 2$

$y(0) = 2 \rightarrow 2 = ae^0 + be^0 + 0 \cdot e^0 + 2 \cdot 0 + 1 \rightarrow 2 = a + b + 1 \rightarrow 1 = a + b$

$y'(0) = 5 \rightarrow 5 = 2ae^0 + 3be^0 + 3 \cdot 0 \cdot e^0 + 2 \rightarrow 5 = 2a + 3b + 2 \rightarrow 3 = 2a + 3b$

$1 = b \rightarrow 0 = a$

$y(x) = e^{3x} + xe^{3x} + 2x + 1 \; ; x \in \mathbb{R}$

②

$y''' + 9y' = 8\cos(x) - 9$

a)

$\lambda^3 + 9\lambda = 0$

$\lambda \cdot (\lambda^2 + 9) = 0$

$\lambda = 0 \qquad \lambda^2 + 9 = 0$

$\lambda = \pm 3i$

$\longmapsto \underline{y_h(x) = a + b\sin(3x) + c\cos(3x)}$

b) $b(x) = 8\cos(x) - 9$

$8\cos(x) \xrightarrow{\text{zsfkl.}} A\cos(x) + B\sin(x) \xrightarrow{\lambda = \pm i}$ bez. kor. $\to A\cos(x) + B\sin(x)$

$-9 \xrightarrow{\text{zsfkl.}} Cx + D \xrightarrow{\lambda = 0}$ bez. kor. $\to Cx + D$

$y_p = A\cos(x) + B\sin(x) + Cx + D$

$y_p' = -A\sin(x) + B\cos(x) + C$

$y'' = -A\cos(x) - B\sin(x)$

$y''' = A\sin(x) - B\cos(x)$

$L = [A\cos(x) + B\sin(x) + Cx + D]''' + 9[A\cos(x) + B\sin(x) + Cx + D]' = A\sin(x) - B\cos(x) - 9A\sin(x) + 9B\cos(x) + 9C = -8A\sin(x) + 8B\cos(x) + 9C \overset{?}{=} 8\cos(x) - 9$

$\begin{bmatrix} -8A = 0 \\ 8B = 8 \\ 9C = -9 \end{bmatrix} \to \begin{matrix} A = 0 \\ B = 1 \\ C = -1 \end{matrix} \longrightarrow y_p = \sin(x) - x$

$\underline{y(x) = a + b\sin(3x) + c\cos(3x) + \sin(x) - x \; ; \; x \in \mathbb{R}}$

$y'(x) = 3b\cos(3x) - 3c\sin(3x) + \cos(x) - 1$

$y''(x) = -9b\sin(3x) - 9c\cos(3x) - \sin(x)$

$y(0) = 13 \to 13 = a + b\sin(0) + c\cos(0) + \sin(0) - 0 \to 13 = a + c$

$y'(0) = 0 \to 0 = 3b\cos(0) - 3c\sin(0) + \cos(0) - 1 \to 0 = 3b + 1 - 1 \to b = 0$

$y''(0) = 0 \to 0 = -9b\sin(0) - 9c\cos(0) - \sin(0) \to 0 = -9c \to c = 0$

$\left.\right\} \; a = 13, \; b = 0, \; c = 0$

$\underline{\underline{y(x) = 13 + \sin(x) - x \; ; \; x \in \mathbb{R}}}$